

Lab 1-2

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The algorithm for constructing a reflexive transitive ~~clo~~ closure, firstly is to add reflexive pairs (a, a) ... for every $a \in A$ (A is the ~~set~~ tuple in program). Then enumerate all ~~is~~ length- i sequence $(b_1, b_2, \dots, b_i)$ for $i = 2 \dots |A|$. If every adjacent pair $(b_j, b_{j+1}) \in R$, insert its endpoint pair ~~(b_1, b_i)~~ (b_1, b_i) into R^* .

Lab1-2 Code

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```
1. from itertools import product
2. from typing import Hashable, Tuple, Set, Iterable
3. import random
4.
5. Pair = Tuple[Hashable, Hashable]
6.
7. def reflexive_transitive_closure(A: Iterable[Hashable], R: Set[Pair]) -> Set[Pair]:
8.     A = tuple(A)
9.     n = len(A)
10.    R = set(R)
11.    Rstar: Set[Pair] = set()
12.
13.    for a in A:
14.        Rstar.add((a, a))
15.
16.    for i in range(2, n+1):
17.        for seq in product(A, repeat=i):
18.            flag = True
19.            for j in range(i-1):
20.                if (seq[j], seq[j+1]) not in R:
21.                    flag = False
22.                    break
23.            if flag:
24.                Rstar.add((seq[0], seq[-1]))
25.
26.    return Rstar
27.
28. def rand_R(n: int, edge_prob: float, seed: int = 42, include_self_in_R: bool = False):
29.     rand = random.Random(seed)
30.     A = list(range(n))
31.     R: Set[Pair] = set()
32.     for x in A:
33.         for y in A:
34.             if (x == y) and not include_self_in_R:
35.                 continue
36.             if rand.random() < edge_prob:
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37.         R.add((x,y))
38.     return A,R
39.
40. if __name__=="__main__":
41.     n=3
42.     p=0.5
43.
44.     A, R = rand_R(n, p)
45.
46.     print("A =", A)
47.     print("R (random edges) =", sorted(R))
48.
49.     Rstar= reflexive_transitive_closure(A, R)
50.
51.     print("\nR* by enumeration =", sorted(Rstar))

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Output

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[Running] python -u "/Users/profighted/Desktop/大四上课程PPT/计算与复杂性/作业/Lab1/lab1-2.py"
A = [0, 1, 2]
R (random edges) = [(0, 2), (1, 0), (1, 2)]

R* by enumeration = [(0, 0), (0, 2), (1, 0), (1, 1), (1, 2), (2, 2)]

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[Running] python -u "/Users/profighted/Desktop/大四上课程PPT/计算与复杂性/作业/Lab1/lab1-2.py"
A = [0, 1, 2, 3, 4]
R (random edges) = [(0, 2), (0, 3), (0, 4), (1, 4), (2, 1), (2, 3), (3, 0), (3, 1), (4, 0), (4, 3)]

R* by enumeration = [(0, 0), (0, 1), (0, 2), (0, 3), (0, 4), (1, 0), (1, 1), (1, 2), (1, 3), (1, 4), (2, 0), (2, 1), (2, 2), (2, 3), (2, 4), (3, 0), (3, 1), (3, 2), (3, 3), (3, 4), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]

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[Running] python -u "/Users/profighted/Desktop/大四上课程PPT/计算与复杂性/作业/Lab1/lab1-2.py"
A = [0, 1, 2, 3, 4, 5, 6, 7]
R (random edges) = [(0, 2), (1, 0), (1, 3), (1, 6), (2, 6), (3, 6), (3, 7), (5, 7), (6, 2)]

R* by enumeration = [(0, 0), (0, 2), (0, 6), (1, 0), (1, 1), (1, 2), (1, 3), (1, 6), (1, 7), (2, 2), (2, 6), (3, 2), (3, 3), (3, 6), (3, 7), (4, 4), (5, 5), (5, 7), (6, 2), (6, 6), (7, 7)]

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